

Continuous Assessment Test (CAT) – I AUGUST 2026

Programme	M. Tech CSE and its specializations	Semester	:	Fall 2026 – 2027
Course Code & Course Title	MACSE501 - Data Structures and Algorithm Analysis	Class Number	:	CH2026270103462
Faculty	Dr. VIJAYALAKSHMI A	Slot	:	BI+TBI
Duration	90 Minutes	Max. Mark	:	50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Use graph sheets supplied from the exam cell as necessary
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1	a.	Prove that $f(n) = 100n + 5 \neq \Omega(n^2)$ [4 Marks]	10	1	K1
	b.	<pre> void XXX(int n) { int i, j, k, count = 0; for(i = n/2; i <= n; i++) { for(j = 1; j + n/2 <= n; j++) { for(k = 1; k <= n; k = k * 2) count++; } } printf("%d", count); } </pre> i. Determine the time complexity of the function XXX() using asymptotic notation (Big-O). Justify your answer. [4 Marks] ii. For the input $n = 5$ (where n is a positive integer), determine and write the output produced by the function XXX(). [2 Marks]			
2	a.	Formulate the recurrence relation to compute the time complexity of the following recursive function YYY(). [5 Marks] <pre> void YYY(int n) { if (n < 2) return; else { counter = 0; </pre>	10	1	K3

		<pre> for(i = 1; i <= 8; i++) YYY(n/2); for(i = 1; i <= n*n*n; i++) counter = counter + 1; } </pre>			
	b.	Solve the following recurrence relation using backward substitution method. [5 Marks] $T(n) = T(n-1) + T(n-2) + 1$			
3		Given a stack S containing n integers, write a pseudocode to determine whether every successive pair of elements in the stack consists of consecutive integers. i.e two integers are considered consecutive if their difference is 1. The consecutive integers may be in either increasing or decreasing order. If the stack contains an odd number of elements, ignore the topmost element, as it does not form a pair. Your pseudocode must preserve the original contents and order of the stack after execution and must use only stack operations. Example: 1 Input: Stack [8, 9, 15, 14, -5, -4, 22, 21, 7] Output: True Example: 2 Input: Stack [9, 7, 4, 5, 2, 1] Output: False	10	1	K3
4		Consider an array $A = [a_0, a_1, \dots, a_{n-1}]$ containing n elements and k be any positive integer. Write a pseudocode to sort and print the elements of A in ascending order only when the index of elements in A are divisible by k and other elements need not be sorted and place it in same index position. Also, derive the time complexity of your pseudocode using asymptotic notation. Example: Input: $A = [1, 11, 5, 6, 3, 7, 9, 13, 2]$ and $k=2$ Output:	10		K5

		A=[1, 11, 2, 6, 3, 7, 5, 13, 9]			
5		Consider a circular singly linked list P containing n nodes, where each node stores an integer value. i. Write a pseudocode to create the circular singly linked list P. [4 Marks] ii. Given a positive integer k ($1 \leq k \leq n$), write a pseudocode to find and print the maximum product obtained by multiplying any k consecutive nodes in the circular singly linked list. [6 Marks] Example Input (P) : Head $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4 $\rightarrow$ 5 $\rightarrow$ 6 $\rightarrow$ Head $k = 3$ Output: 120 ($6 \times 2 \times 3$)	10	1	K6

*****All the best *****